

# Dr. Deniz Berfin Karakoc

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## RESEARCH INTERESTS

|                 | <u>Methodology</u>             | <u>Application</u>            |
|-----------------|--------------------------------|-------------------------------|
| Optimization    | Data analytics                 | Agriculture and food systems  |
| Network science | Geographic information science | Transportation infrastructure |

My research addresses the 21<sup>st</sup> century challenges against cyber-physical-social systems. I am particularly interested in *enhancing the multi-dimensional resilience and sustainability of agri-food systems and their supporting critical infrastructure*. Please see my [Google Scholar](#) for more details.

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## EDUCATION

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| Aug 2019 – Aug 2024   | <b>Ph.D., Civil and Environmental Engineering</b><br>Sustainable and Resilient Infrastructure Systems<br>University of Illinois Urbana-Champaign, USA |
| Aug 2017 – May 2019   | <b>M.S., Industrial and Systems Engineering</b><br>University of Oklahoma, USA                                                                        |
| Aug 2017 – May 2019   | <b>Grad.Cert., Industrial and Systems Engineering</b><br>University of Oklahoma, USA                                                                  |
| Sept 2012 – July 2016 | <b>B.S., Industrial Engineering</b><br>Bilkent University, TURKEY                                                                                     |

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## PROFESSIONAL EXPERIENCE

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| Aug 2024 – Present  | <b>Assistant Professor</b> , Arizona State University<br>School of Computing and Augmented Intelligence                    |
| Aug 2019 – Aug 2024 | <b>Graduate Research Assistant</b> , University of Illinois Urbana-Champaign                                               |
| Aug 2017 – May 2019 | <b>Graduate Research Assistant</b> , University of Oklahoma<br><b>Graduate Teaching Assistant</b> , University of Oklahoma |

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## AWARDS and HONORS

**Nature Food Cover** – Volume 4 Issue 7, July 2023

*Structural chokepoints determine the resilience of agri-food supply chains in the United States*

**Dissertation Completion Fellowship** – University of Illinois Urbana-Champaign Graduate College, 2023-2024

*The only awarded senior Ph.D. candidate from the Department of Civil and Environmental Engineering*

**CEE Rising Stars** – Massachusetts Institute of Technology, 2023

**ACDIS Summer Fellowship** – University of Illinois Urbana-Champaign College of Liberal Arts & Sciences, 2023

**Best Thesis Award** - University of Oklahoma School of Industrial and Systems Engineering, 2019

*The only awarded senior M.S. student from the School of Industrial and Systems Engineering*

**14<sup>th</sup> Industrial Engineering Senior Design Project Contest and Fair Winner** - Bilkent University, 2016

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## PUBLICATIONS

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### Peer-reviewed Journal Articles (published)

13. Zhang R., Arnav, A., **Karakoc D. B.** and Konar M. 2025. [Compound shocks to agri-food supply chain transport and trade in the United States](#). *Frontiers in Sustainable Food Systems*, 9: 1661492. **(Impact Factor: 3.1)**
12. **Karakoc, D. B.**, and Konar M. 2025. [Multimodal and multicereal grain transportation patterns in the United States across cost, emission, and adaptability optimization scenarios](#). *Environmental Science & Technology*, 59 (34): 18156-18166. **(Impact Factor: 11.4)**
11. **Karakoc, D. B.**, and Konar M. 2025. [Trade-off between resilience, sustainability, and cost in U.S. agri-food transportation infrastructure](#). *Nature Food*, 1-9. **(Impact Factor: 23.6)**
10. **Karakoc, D. B.**, and Konar M. 2024. [Optimization of national grain imports to balance risk and return: A portfolio theory approach](#). *Environmental Research: Food Systems*, 1: 011001.
9. **Karakoc, D. B.**, Konar M., Puma M.J. and Varshney L.V. 2023. [Structural chokepoints determine the resilience of agri-food supply chains in the United States](#), *Nature Food*, 4: 607–615. **(Impact Factor: 23.6) (Nature Food July 2023 Cover)**
8. Pandit, A., **Karakoc D. B.** and Konar M. 2023. [Spatially detailed agricultural and food trade between China and the United States](#), *Environmental Research Letters*, 18: 084031. **(Impact Factor: 6.947)**
7. **Karakoc, D. B.**, Barker K. and González A. 2023. [Analyzing the tradeoff between vulnerability and recoverability investments for interdependent infrastructure networks](#), *Socio-Economic Planning Sciences*, 87: 101508. **(Impact Factor: 4.641)**
6. Maraqa, S. N., **Karakoc D. B.**, Ghorbani-Renani N., Barker K. and González A. 2022. [Project schedule compression for the efficient restoration of interdependent infrastructure systems](#), *Computers & Industrial Engineering*, 170: 108342. **(Impact Factor: 7.18)**
5. Wang, J. **Karakoc D. B.** and Konar M. 2022. [The Carbon Footprint of Cold Chain Food Flows in the United States](#), *Environmental Research: Infrastructure and Sustainability*, 2: 021002.
4. **Karakoc, D. B.**, Wang J. and Konar M. 2022. [Food flows between counties in the United States from 2007 to 2017](#), *Environmental Research Letters*, 17: 034035. **(Impact Factor: 6.947)**
3. **Karakoc, D. B.** and Konar M. 2021. [A Complex Network Framework for the Efficiency and Resilience Trade-off in Global Food Trade](#), *Environmental Research Letters*, 16: 105003. **(Impact Factor: 6.947)**
2. **Karakoc, D. B.**, Barker K., Zobel C. and Almoghathawi Y. 2020. [Social-Vulnerability and Equity Perspectives on Interdependent Infrastructure Network Component Importance](#), *Sustainable Cities and Society*, 57: 102072. **(Impact Factor: 10.969)**
1. **Karakoc, D. B.**, Almoghathawi Y., Barker K., González A. and Mohebbi S. 2019. [Community-Resilience Driven Restoration Model for Interdependent Infrastructure Networks](#), *International Journal of Disaster Risk Reduction*, 38: 101228. **(Impact Factor: 4.842)**

### Peer-reviewed Conference Articles (published)

3. Gonzalez, E., Grogan P., **Karakoc D. B.**, Mizuyama H., Sato M., Suginochi S. and Zarreh M. 2025. [Modeling and Comparison of U.S. and Japanese Lettuce Supply Chain System Architecture](#). *Advances in Production Management Systems. Cyber-Physical-Human Production Systems: Human-AI Collaboration and Beyond*. APMS. DOI: 10.1007/978-3-032-03542-4\_18
2. **Karakoc, D. B.**, Barker K. and Almoghathawi Y. 2019. [Interdependent Infrastructure Network Restoration Optimization Problem from Community and Spatial Resilience Perspective](#), In INOC 2019 *Proceedings of International Network Optimization Conference, Avignon, France*.
1. Barker, K., **Karakoc D.B.** and Almoghathawi Y. 2018. [Interdependent Infrastructure Network Restoration Problem from Community-Resilience Perspective](#), In ESREL 2018 *Proceedings of European Safety and Reliability Conference, Trondheim, Norway*.

#### Peer-reviewed Journal Articles (submitted and in preparation)

5. Cai Y., Polat, S. U., and **Karakoc, D. B.** 2025. Evaluation of low-income and food-deficit nations' grain imports to balance reliability and cost. Second round-of-revision in: *Nature Food*.
4. Sardashti, T., Ghozatfar, A., and **Karakoc D. B.** 2025. Uncovering the risk dynamics of staple grain imports: A global and national perspective. Second round-of-revision in: *Nature Communications Earth & Environment*.
3. Kumar, A. and **Karakoc D. B.** 2025. Spatio-temporal food consumption trends: Implications for food security and sustainability in the United States. In preparation.
2. Ghozatfar, A. and **Karakoc D. B.** 2025. Revealing the export share determinants and future leading nations of agri-food trade. Submitted to: *npj Science of Food*
1. Arnav, A., Zhang R., **Karakoc D. B.** and Konar M. 2025. Shocks to agri-food flows between counties in the United States from 2018 to 2022. Submitted to: *Environmental Research: Food Systems*.

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### ACADEMIC SERVICES

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#### Leadership Services

- Session Chair, INFORMS Annual Meeting 2023 – 2024 – 2025
- Session Convener, AGU Fall Meeting 2023 – 2024 – 2025

#### Outreach Services

- Data production and preparation, “Food flows between U.S. counties in 2007, 2012, and 2017”, [foodflows.org](http://foodflows.org)
- K-12 instructor, “Resilient food supply chains”, CEE Summer Camp 2023, [wyse.engineering.illinois.edu](http://wyse.engineering.illinois.edu)
- Blog post preparation, “Mapping Food Flow Networks and the Food Supply Chain”, [farmdocdaily.illinois.edu](http://farmdocdaily.illinois.edu)

#### Peer-review Services

- **Journals:** Food Security (2025), IEEE Transactions on Automation Science and Engineering (2025), Sustainable Cities and Society (2025), Environmental Research: Food Systems (2025), Nature Communications (2025), Physica Scripta (2024), Frontiers in Sustainable Food Systems (2024), IEEE Transaction on Network Science and Engineering (2024), Earth's Future (2023 - 2025), Journal of Industrial Ecology (2023), Nature Food (2022 - 2025), Nature Scientific Reports (2022), Food Policy (2022), Water Resources Research (2021), Economic System Research (2020), PLoS ONE (2020)
- **Proposals:** USDA Food and Agricultural Sciences National Needs Graduate and Postgraduate Fellowship Grants Program (2024), NSF Division of Social and Economic Sciences Decision, Risk, and Management Sciences Program (2025).

- **Conferences:** INFORMS Annual Meeting Poster Session (2025).

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## SELECTED PRESENTATIONS

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### Conference Talks

10. Spatio-temporal food consumption trends: Implications for food security and sustainability in the U.S., The Institute for Operations Research and the Management Sciences, **INFORMS Annual Meeting 2025**.
9. Mapping agri-food distribution onto real-world transportation infrastructure in the U.S., The Institute for Operations Research and the Management Sciences, **INFORMS Annual Meeting 2024**.
8. Optimization of national grain imports to balance risk vs. return: A portfolio theory approach, The Institute for Operations Research and the Management Sciences, **INFORMS Annual Meeting 2023**.
7. Carbon footprint of cold-chain food flows between counties in the United States, Production and Operations Management Society, **POMS Annual Conference 2023**.
6. Food flows between U.S. counties through time, The American Geophysical Union, **AGU Fall Meeting 2021**.
5. Resilience and efficiency in food trade networks, The American Geophysical Union, **AGU Fall Meeting 2020**.
4. Community-resilience driven trade-off analysis between the pre- and post-event investments for interdependent infrastructure networks, Institute of Industrial and Systems Engineers, **IISE Annual Conference & Expo 2019**.
3. A social-vulnerability driven component importance measure for interdependent infrastructure networks, The Institute for Operations Research and the Management Sciences, **INFORMS Annual Meeting 2018**.
2. Interdependent Infrastructure Network Restoration Problem from Multiple Community-Resilience Approaches, Institute of Industrial and Systems Engineers, **IISE Annual Conference & Expo 2018**.
1. Hot Sales Delivery System Logistics Optimization and Cost Analysis, The Institute for Operations Research and the Management Sciences, **INFORMS Annual Meeting 2016**.

### Invited Talks

6. Unraveling spatio-temporal food supply chain patterns. School of Sustainability Seminar Series, 2025, **Arizona State University**.
5. Unraveling spatio-temporal food supply chain patterns to enhance resilience, sustainability, and equity. School of Agribusiness Seminar Series, 2025, **Arizona State University**.
4. Resilient, sustainable, and equitable agri-food systems across spatial scales. Hydro-systems Seminar Series, 2025, **Arizona State University**.
3. Community resilience perspective on interdependent infrastructure network restoration planning and component importance analysis. Disaster Resilience Analytics Center Seminar Series, 2022, **Wichita State University**.
2. Food supply chain bottlenecks. 2022, **The MITRE Corporation**.
1. Network analysis of food supply chains. Sustainability & Resistant Infrastructure Systems Seminar Series, 2021, **University of Illinois Urbana-Champaign**.

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## MEDIA COVERAGE

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Physics Org: “**Keeping America’s cereal bowl full: Optimizing grain transport to balance cost, carbon and resilience**”, by Kelly deVos, 09 September 2025. <https://phys.org/news/>

ASU Full Circle: “**Transporting food from farm to table**”, by Kelly deVos, 07 March 2025.

<https://fullcircle.asu.edu/research/>

American Council on Science and Health: “**From Farm to Fork: Our Food Supply Chain**”, by Chuck Dinerstein, 01 August 2023. <https://www.acsh.org/news/>

New Food Magazine: “**How resilient are US food supply chains?**” by Grace Galler, 24 July 2023.

<https://www.newfoodmagazine.com/news/>

Nature Food News & Views: “**Logistics hubs hold food supply chains together**”, by Graham K. MacDonald, 20 July 2023. <https://www.nature.com/articles/>

Department News: “**Researchers illuminate resilience of U.S. supply chains**”, by UIUC Civil and Environmental Engineering. 20 July 2023. <https://cee.illinois.edu/news/>